

Jongyun Hur

📍 Seoul, South Korea ✉️ jongyun.jh@gmail.com ☎️ +82-10-3900-4036 🌐 Website in Jongyun Hur
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Research Interests

My primary research interest lies in computer architecture and hardware-software co-design for emerging computing systems. I enjoy exploring how hardware and software work together to create next-generation architectures and hardware accelerators. My goal is to enhance system efficiency and minimize energy consumption through effective hardware-software optimization.

Education

Soongsil University Seoul, South Korea
B.S. in Artificial Intelligence Convergence Mar. 2022 – Feb. 2028 (Expected)

- **GPA:** Total 3.59/4.5, Major 3.54/4.5 (73.5/133 credits completed)
- **Specialization Track:** Circuit Systems Engineering (Dept. of Next-Generation Semiconductor)

Internship

Potatonet Seoul, South Korea
Machine Learning Engineer Intern Aug. 2025 – Nov. 2025

- Developing VLM-based CCTV monitoring system on Jetson Orin Nano for real-time fire detection and escalator safety monitoring. (In Progress)
- Building an integrated MLOps platform to automate and organize model training workflows across teams, converting research outputs into reusable resources. (In Progress)
- Optimized an end-to-end YOLO logo detection pipeline for an A100 GPU server, utilizing Redis/Celery parallel workers to maximize GPU utilization and memory efficiency.

Project Experience

XAI-based Explainable Military Decision Support System Seoul, South Korea
TEAM LEADER Jun. 2024 – Oct. 2024

- Presented an XAI-based military decision support system at 2024 Republic of Korea Army AI Idea Competition.
- Developed command decision support prototype using SHAP, LSTM, and Sentence-BERT for explainable tactical recommendations in Korean military domain.
- Implemented Military NER tagging and prompt engineering techniques for Korean language understanding in tactical decision workflows.

Multi-threaded Lock Implementation and Performance Analysis Seoul, South Korea
TEAM MEMBER Oct. 2023 – Dec. 2023

- Team project for course Computer Architecture of 2023-Fall Semester.
- Implemented spinlock and ticket lock using ARMv8 LDAXR/STLXR atomics and validated correctness under multi-threaded contention via shared-array tests.
- Built Pthreads benchmarks to measure runtime and throughput across multiple thread configurations.
- Analyzed performance trade-offs, comparing the fairness and scalability of both lock implementations under varying thread counts

Real-time Voice Phishing Detection System using Deep Learning

TEAM LEADER, PRESENTER

Seoul, South Korea
Sep. 2023 – Nov. 2023

- Presented project at 2023 KISA AI+Security Idea Competition.
- Designed and implemented an ensemble model combining text-based NLP (BERT) and audio-based analysis (MFCC) for real-time phishing detection.
- Built an end-to-end ML pipeline including data augmentation, feature extraction, model training, and deployment architecture.

Awards

2nd Place, Soongsil University AI Convergence Competition

Soongsil University

Skills

- Programming: C++, C, Python, SQL, Verilog
- Deep Learning: Scikit-learn, PyTorch, Keras, TensorFlow

Other

Languages

- Korean (Native)
- English (Intermediate Proficiency, TOEIC: 770, *TOEFL score pending*)

Certifications:

- 2025.06 - Information Processing Industrial Engineer
- 2025.05 - TOPCIT Level 3
- 2024.10 - KAIST Military ICT Intensive Program
- 2024.06 - ADSP
- 2024.06 - SQLD